

Unit 7, Lesson 6: Area of Composite Figures



Benchmark

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.



Learning Targets

- ☐ I can find the area of composite figures.
- ☐ I can find the value of a missing dimension for a quadrilateral or a composite figure.



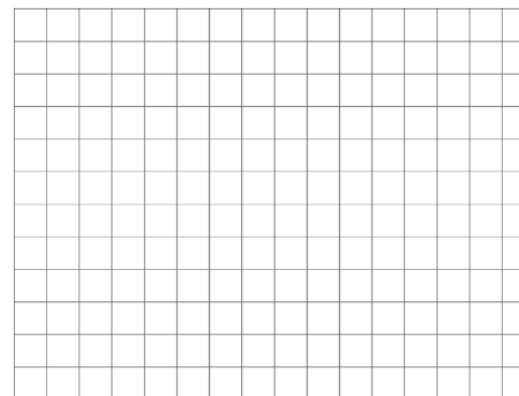
7.6.1 Warm-Up: Closest to Six

- Use the digits 0 to 9, at most one time each, to fill in ordered pairs for all three points, such that the area of Triangle ABC is closest to 6 square units.

A (,)

B (,)

C (,)



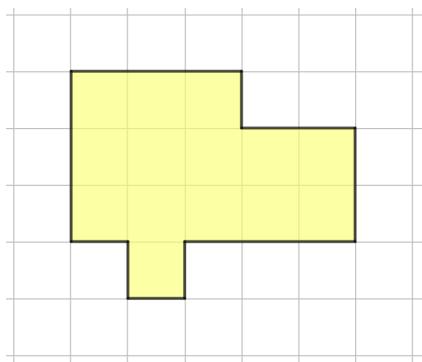


7.6.2 Guided Instruction: Area of Composite Figures



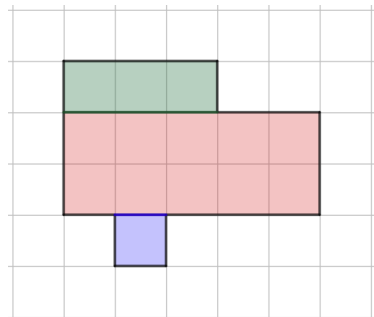
A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

Jackson and Danna were instructed to find the area of the composite figure below. The length of each square is 1 centimeter (cm).

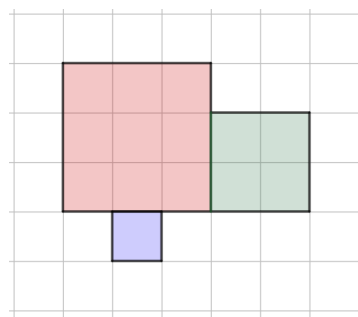


Both Jackson and Danna decomposed the figure, but their teacher noticed they each decomposed it differently.

Jackson



Danna



1. Complete the table about each student's work. Include units.

Student	Area of Red Rectangle	Area of Blue Rectangle	Area of Green Rectangle
Jackson			
Danna			

2. To find the area of the composite figure, find the sum of the areas of the rectangles for each student. Include units.

	Area of the Composite Figure
Jackson	
Danna	

3. Explain why both students got the same area of the composite shape.

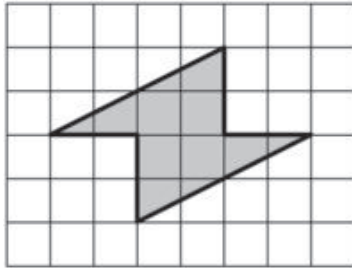


The **area of a composite figure** is the sum of the areas of the figure's decomposed parts.



7.6.3 Guided Practice: Area of Composite Figures

1. A composite shape is shown. The length of each square is 1 inch (in).



- Draw lines to decompose the figure into rectangles and/or triangles.
- Find the area of each rectangle and/or triangle. Include units.
- Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.
- Find a classmate who decomposed the shape differently than you did. Describe their process.

2. Thomas has decided to put grass sod in his backyard. A drawing of his backyard is shown.



How much sod does Thomas need to buy?



7.6.4 Collaboration: Finding a Missing Value

- Complete the table by filling in the blanks for the following math facts.

If...	Then...
$40 \div 8 = \underline{\hspace{2cm}}$	$8 \times \underline{\hspace{2cm}} = 40$
$27 \div 3 \div \frac{1}{2} = \underline{\hspace{2cm}}$	$\frac{1}{2} \times \underline{\hspace{2cm}} \times 3 = 27$
$8 \div \frac{8}{7} = \underline{\hspace{2cm}}$	$\underline{\hspace{2cm}} \times \frac{8}{7} = 8$

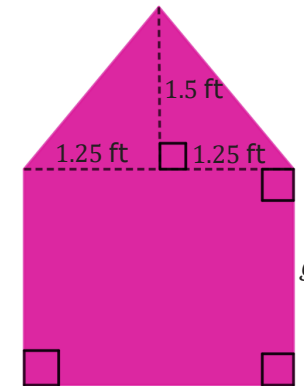
- With your partner, use the previous math facts to find the missing dimensions of the quadrilaterals. All measurements are in centimeters (cm).

Area = 40 cm^2	Area = 27 cm^2	Area = 8 cm^2
$k = \underline{\hspace{2cm}}$	$m = \underline{\hspace{2cm}}$	$x = \underline{\hspace{2cm}}$



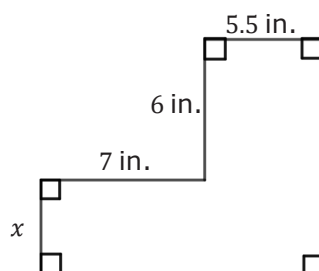
7.6.5 Guided Practice: Finding the Value of the Missing Dimension

- Xavier is painting the back wall of a dollhouse. He knows the back wall of the dollhouse has an area of $6.875 \text{ square feet (ft}^2\text{)}$.



- What is the area of the triangular top? Include units.
- What is the area of the rectangular base? Include units.
- What is the height, g , of the rectangular base? Include units.

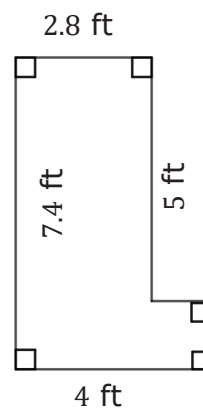
2. The area of the composite figure is 84 square inches (in^2). Find the missing dimension. Include units.



7.6.6 Your Turn!

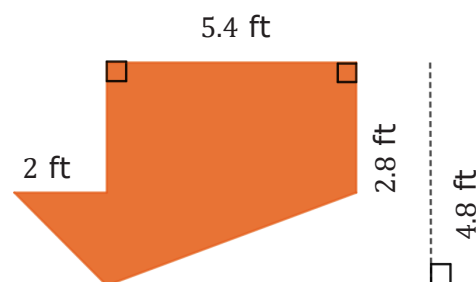
For each composite figure, find the area, in square feet (ft^2). Show your work.

1.



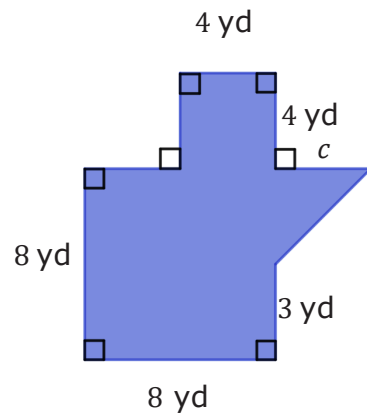
Area = _____

2.



Area = _____

3. The area of the composite figure is 87.5 square yards (yd^2). Find the value of the missing dimension. Include units.



Not drawn to scale



7.6.7 Wrap-Up: Reflection

Complete each statement based about today's lesson.

1. I am still unsure about ...

2. One question I still have about today's lesson is ...
